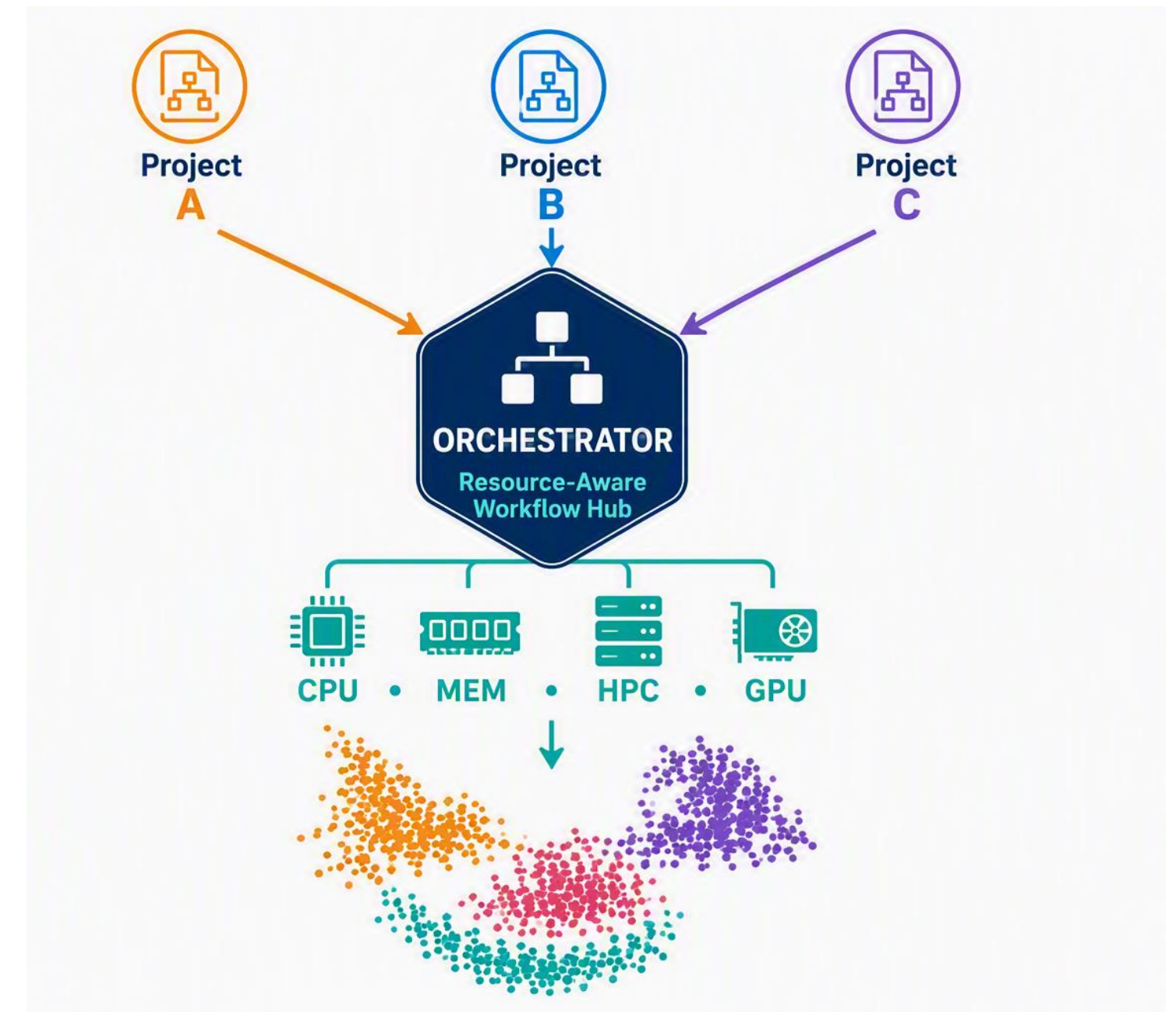


ResMetis

Resource-Aware Multi-Project Orchestrator for Single-Cell Analysis

St. Jude BioHackathon 2026 Project
Kids26 Team 15



Today's Agenda Items

1. St. Jude BioHackathon 2026 Project Kids26 Team 15
2. Confirm access to tools, apps etc
3. Project: scope, timeline, responsibilities
4. Tools and resources



Kids26 Team 15 – let's get to know each other!

- **Antonia Chroni, PhD, Technical Contributor and Project Lead.** Lead Bioinformatics Research Scientist, Department of Developmental Neurobiology, St. Jude Children's Research Hospital.
- **Sarthak Sunil Dhanke, Technical Contributor.** Research Assistant, University of Chicago.
- **A. S. M. Tanjim Hassan, Technical Contributor.** Undergraduate Student, University of Dhaka.
- **Rachana Pandey, Technical Contributor.** Graduate Research Assistant, University of Minnesota, Twin Cities.
- **Rojina Sapkota, Data Explorer, Technical Contributor.** Bioinformatics Analyst, St. Jude Children's Research Hospital.
- **Jason Vu, Technical Contributor.** St. Jude Children's Research Hospital.
- **Lindsey Warren, Technical Contributor.** Enterprise Architect. International Paper.

For more information, see [BioHackathon list](#).



Kids26 Team 15 – pending tasks due 09/04/2026

- In Person Attendance
- Accepted your GitHub invitations?
- Added in the Slack channel of team15? This will be used for our communication moving forward.
- Added in the GitHub [team](#) and repo: <https://github.com/stjude-biohackathon/KIDS26-Team15>
- Technical skills - go through them on excel file with everyone
- Explain pairs and responsibilities
 - GitHub experience
 - sc-rna-seq data analysis experience

For more information, see [dropbox excel file](#).



GitHub repo: <https://github.com/stjude-biohackathon/KIDS26-Team15>

stjude-biohackathon / KIDS26-Team15

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Issues 2

Pull requests 1

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

KIDS26-Team15

Public

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Fork 0

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main 2 Branches 0 Tags

Go to file T

Add file

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AntoniaChroni

Merge pull request #3 from stjude-biohackathon/add-forked-repo

a5f9c9e · last week

5 Commits

.github	Rm codeowners	last week
analyses	Add forked repo	last week
docs	Initial commit	3 weeks ago
project-management	Initial commit	3 weeks ago
.all-contributorsrc	Initial commit	3 weeks ago
.gitignore	Initial commit	3 weeks ago
LICENSE.md	Initial commit	3 weeks ago
README.md	Initial commit	3 weeks ago

README MIT license

Biohackathon Project Template

This repository is a starting point for a three-day team project. This repository is populated with a starting template for team organization and planning. Use it to plan, build, and document work. Please adjust this repository to suit the needs of your team.

About

No description, website, or topics provided.

Readme

MIT license

Activity

Custom properties

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 2

ramsallaq

Ramzi Alsallaq

AntoniaChroni

Antonia Chroni



GitHub repo: <https://github.com/stjude-biohackathon/KIDS26-Team15>

- [docs](#) – resources and tools we will use during the event – go through it and let me know if you have any questions by **09/10/2026**
- [project-management](#) – follow-up with you by **09/10/2026**.
 - <https://github.com/stjude-biohackathon/KIDS26-Team15/pull/7>



Tools and resources

- [vm-local-only-setup.md](#) – follow-up with you by 09/10/2026. Everyone will have to test access to the VM and dry-test software/tools as needed by 09/15/2026 to allow for troubleshooting.
 - Personal laptops
 - HPC-StJude User



BioHackathon Project Timeline

- **Day 1 | 4 hours:** Project Setup and Planning
 - Align on project goals and expected demo outcome
 - Define team roles and responsibilities
 - Finalize the technical approach
 - Set up the development environment and begin implementation
- **Day 2 | 8 hours:** Core Development and Testing
 - Implement the main project components
 - Integrate team contributions
 - Test and troubleshoot the workflow
 - Document key decisions, limitations, and results
 - Start preparing the demo
- **Day 3 | 4 hours:** Finalize, Demo, and Present
 - Complete final testing and documentation
 - Prepare the demo and required 3-slide presentation
 - Select 2 to 4 team members to present
 - Practice the 2-minute lightning presentation



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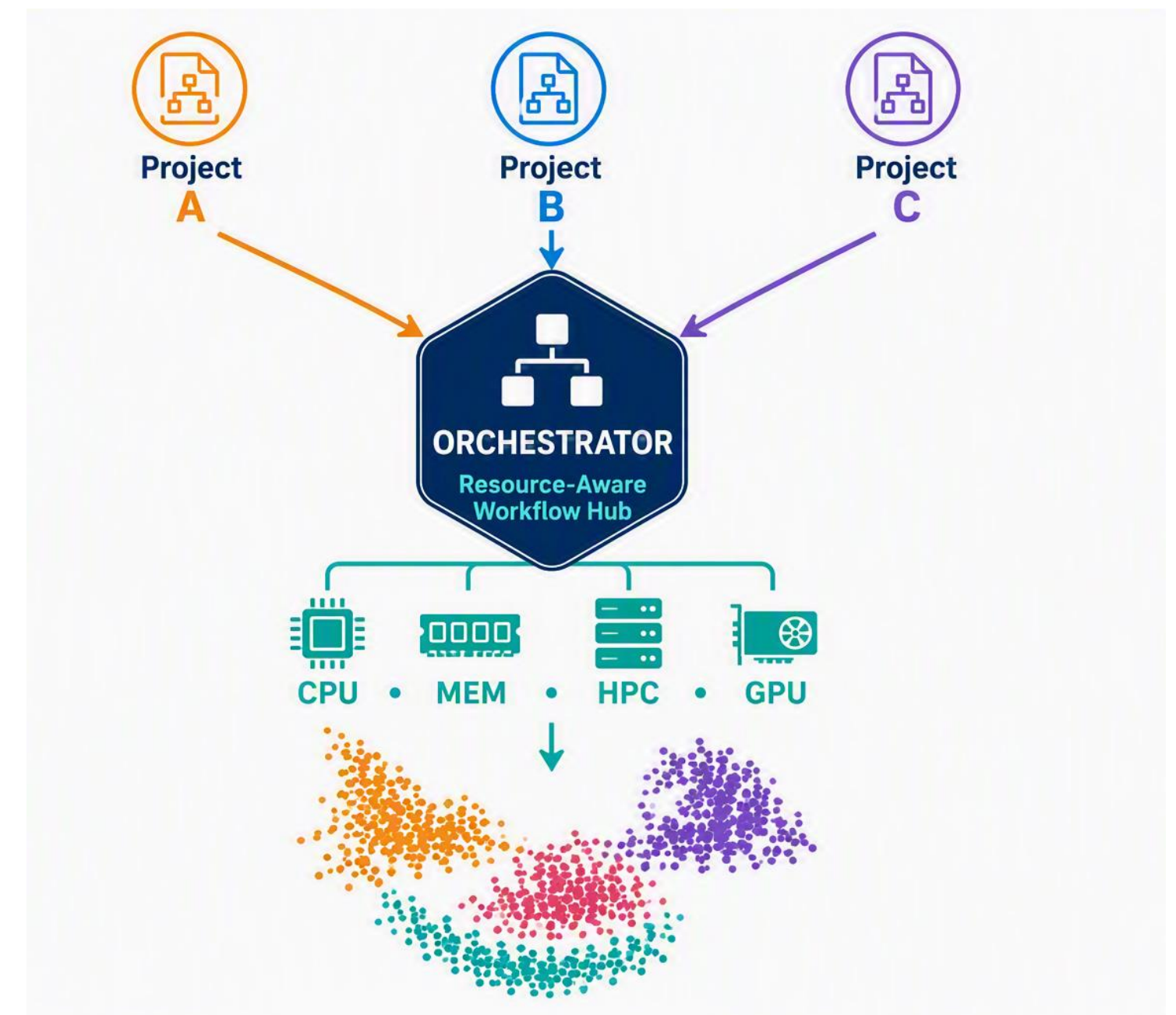


- **3:00–4:00 PM | Project Demo**
 - Deliver a 2-minute lightning presentation to the judges
 - Move to the assigned room for the project demonstration
 - Answer questions and receive feedback
- **4:00–6:00 PM | Demo Lightning Session**
 - Present what the project was about and what was accomplished
 - Judges provide additional feedback



ResMetis: Smarter Resources. Faster Workflows. Scalable Science. 🚀 🧬

- **Resource-Aware + Metis (Greek goddess of wisdom and strategy)**
- Wisdom-Driven Resource Orchestration for Single-Cell Analysis
- Inspired by Metis (Μῆτις), the Greek goddess of wisdom and strategic planning, ResMetis intelligently coordinates computational resources to optimize and scale multi-project single-cell workflows.

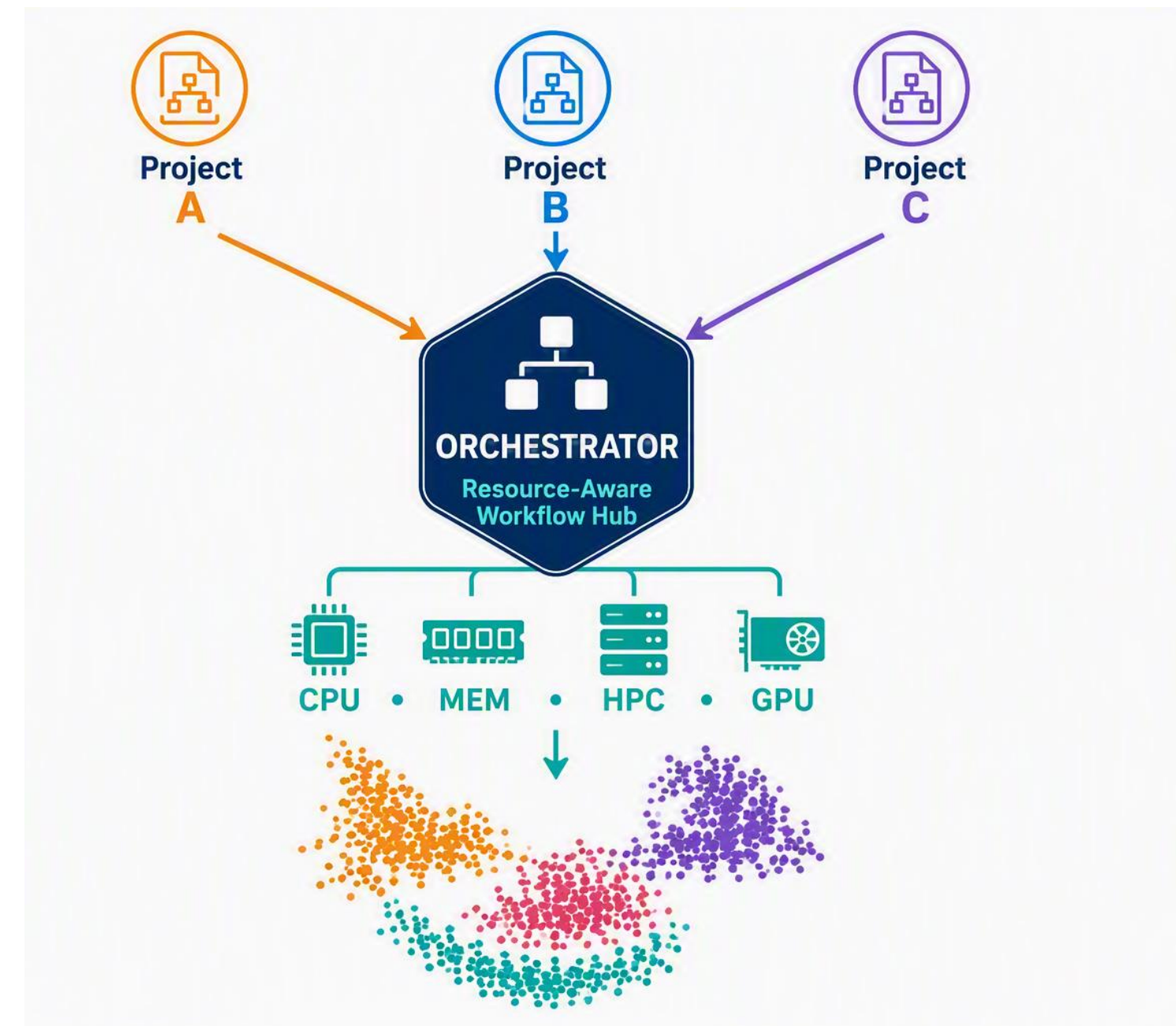




Any ideas are on the name, logo, or other are welcome!

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Current Workflow Landscape and Key Limitations

1. Current Pipeline Workflow

- Demonstrate the [sc-rna-seq-snap pipeline](#) through GitHub
- Focus on one or two representative modules:
 - Upstream analysis
 - Integrative analysis

2. Resource Allocation Challenges

- LSF resource requests are based on pipeline defaults and may not be optimal for every project
- Current sc-rna-seq-snap defaults are designed for approximately:
 - 8 samples
 - 50,000 cells
- Projects with different dataset sizes or algorithmic requirements may encounter resource bottlenecks
- Individual steps may fail when memory, CPU, or runtime requirements exceed the default allocations
- Resource requirements are difficult to predict because they depend on the dataset and the single-cell algorithms used



Current Workflow Landscape and Key Limitations

3. Current Configuration Approach

- Resource parameters are defined as global values in YAML configuration files
- Future direction: use an R package to support more adaptive, project-specific resource estimation

4. Pipeline Execution Options

a) Run a single project or pipeline

- Module-level execution:
root_dir/analyses/<module_name>/lsf-script.txt
- Full-pipeline execution:
root_dir/launch_full_pipeline.sh

b) Run multiple projects

- launch_project_multiple.sh

c) Run multiple pipelines

- Not currently supported

5. Demonstration Dataset. Use the **Victoria Knockout** project as the initial test case





Lightning presentation

3-slide template (required by organizers):

Slide	Owner	Content
1 — Problem	Track C	Manual LSF wiring: analyst time, fixed resources, error-prone bash flags
2 — Solution & what we built	Track A + B	YAML toggles + one-command launcher; test evidence (T1–T13)
3 — Results & impact	Track C + D	ROI table (T17–T18), limitations, next steps

2-minute lightning script:

- 1. Problem in one sentence — why manual LSF wiring hurts analysts (20 s)
- 2. What we tested and what passed — YAML toggles, auto-scaling, one-command launch (60 s)
- 3. Key result + ask — ROI headline, live demo teaser (40 s)

Problem: Manual LSF wiring slows single-cell analysis

KIDS26 Team 15 — Resource-Aware SNAP + Sprocket Orchestrator

- Running SNAP downstream today means manual LSF jobs: separate bsub calls, hand-wired dependencies, and per-module resource guessing
- Analysts spend ~6–8 hours per run on setup, retries, and YAML edits (lsf-script.txt, future.globals)
- Resources are fixed (e.g. 16 CPU / 30 GB) even when the cohort needs less — leading to wasted cores or failures
- No pre-submit validation — errors surface only after jobs are queued on HPC

```
BEFORE:  bsub → wait → edit YAML → retry → 6–8 hr
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Can a resource-aware SNAP + Sprocket orchestrator replace manual LSF wiring for scalable single-cell analysis?

This is an analyst time + cluster waste, not just a tooling upgrade.



Solution & What We Built: One YAML + one command — validated before submit

launch-snap-downstream.sh on Victoria Knockout cohort (4 samples · ~21k cells)

What we built:

Component	What it does
YAML <code>workflow_profile</code>	Turn SNAP modules on/off without editing bash flags
<code>estimate-snap-downstream-resources.R</code>	Auto-scale CPU/memory from Cell Ranger metrics
WDL + Sprocket	<code>sprocket check</code> + <code>validate</code> before any LSF submit
<code>launch-snap-downstream.sh</code>	One command: regen WDL → estimate → validate → <code>--submit</code>

What we will test (3 days):

- **Track A (Rojina + Antonia):** Code review — WDL, launcher, resource estimator
- **Track B (Sarthak):** Test matrix T1–T8 — layout, dry-run, toggles, error handling, scaling
- **Track C (Lindsey + Rachana):** Test log, PR reviews, troubleshooting docs, ROI tables
- **Track D (Tanjim + Antonia):** HPC golden-path submit (T13–T16), biology QC, demo prep

Evidence already in hand:

- Upstream completed: **74 min**, 8 CPU, **24 GB peak** (vs 16 CPU legacy)
- Integrative (Harmony): **15 min**, 10 CPU, **10 GB peak**
- Analyst setup: **~15 min** vs **~2–4 hr** with manual LSF

Speaker note (Tracks A + B): Emphasize “test, don’t just build” — success = logged pass/fail in `results/hackathon-test-results.md`.

Results & Impact: ROI, limitations, and next steps

Results we aim to show judges (Day 3, 3:00–6:00 PM):

Metric	Legacy manual	Our orchestrator
Analyst time / launch	~6–8 hr	~15 min
CPU reservation	16 cores (fixed)	8 cores (auto-scaled)
Pre-submit validation	None	sprocket validate
Multi-module testing	Re-chain bash + bsub	Toggle YAML

Estimated impact (Victoria Knockout):

- ~50% core reservation savings (8 vs 16 CPU)
- ~2–3 hr analyst time saved per launch
- Fewer OOM retries via auto-scaled memory + 20% headroom

Limitations (be honest):

- VM cannot submit to LSF — remote testers need HPC handoff (Rojina / Antonia)
- Day-1 Sprocket tuning took ~17 attempts before stable upstream success
- Full upstream Seurat run on HPC is a **bonus**, not required for demo success

Next steps

- **Other SNAP modalities** — extend to automated sc-ATAC, sc-PARSE, and other pipelines (validated here on sc-RNA-seq)
- **Multi-omics — combined workflows**, e.g. sc-RNA-seq + sc-ATAC, under one resource-aware launcher
- **Multiple projects** — parallel cohort runs via snap_multi_project.wdl
- **Production — CI layout check**; rollout to additional SNAP projects at St. Jude

3-day plan:

- Day 1 (4 h): Clone, smoke tests (T1–T2), confirm pairs
 - Day 2 (8 h): Full test matrix, docs, draft this deck
 - Day 3 (4 h + demo): Finalize 3 slides, 2-min lightning talk, live demo for judges
-
- Speaker note (Tracks C + D): Slide 3 is partly aspirational for kickoff — update numbers from hackathon-test-results.md before the judge session.

